



14-848 Cloud Infrastructure

FALL 2026

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Agenda

- Welcome and Introductions
- Focus Areas of this Course
- Course Syllabus & Schedule
- Class Expectations
- Introduction to the Cloud
- Next Steps



Important Course Aspects

- Relevant course topics to today's market and research
- Practical course: 40% theory + 60% practice
- Course focuses on Cloud Infrastructure (compared to development)
- Assignments are implemented on public cloud environment
 - Students will receive access to Google Cloud from the course instructor but all cloud providers are welcome.
- Three practical project options to choose from.
- **Note: Don't take this course if you took 15319/15619/15719.**



Cloud Infrastructure in Today's Market

In recent years, the demand for Cloud Infrastructure jumped significantly.

- The [International Data Corporation \(IDC\)](#) notes that global spending on public cloud services is forecast to surpass \$1 trillion in 2026, marking growth by over 21%. It is expected to double by 2029.
- [Synergy research](#) estimates that spending on cloud infrastructure services jumped by more than \$43 billion from the second quarter of last year to reach \$143 billion.
- [Gartner](#) identifies that by 2029, cloud computing will be absolutely essential if businesses hope to compete, and for government agencies to achieve their mission.
- [Oxylabs Research](#) analyzed 1M jobs posted in 2025 and 2026 and found nearly 42% of them requiring knowledge of at least one major cloud platform.

Focus Areas of this Course

Cloud-related Technologies

Cloud Providers:

- Google Cloud Platform

Big Data Processing Platforms:

- Apache Hadoop
- Apache Spark

LLM Platforms:

- Google AI Studio

DevOps

Deployment Scaling and Orchestration:

- Docker
- Kubernetes

DevSecOps

- Falco

Cloud Infrastructure

- Infrastructure-as-Code
 - Terraform
- Kafka
- Data Centers
- Edge Computing & Fog Computing



Expectations for Incoming Students

- ***You are expected to know Python or are willing to learn it.***
 - A Python recording is released this week for members who need support with Python
- ***You are expected to have a basic understanding of Computer Virtualization.***
 - If you don't know about virtualization, check out this reading:
<https://www.vembu.com/blog/physical-server-vs-virtual-machine-choice-open/>

Instructor and TA Introductions

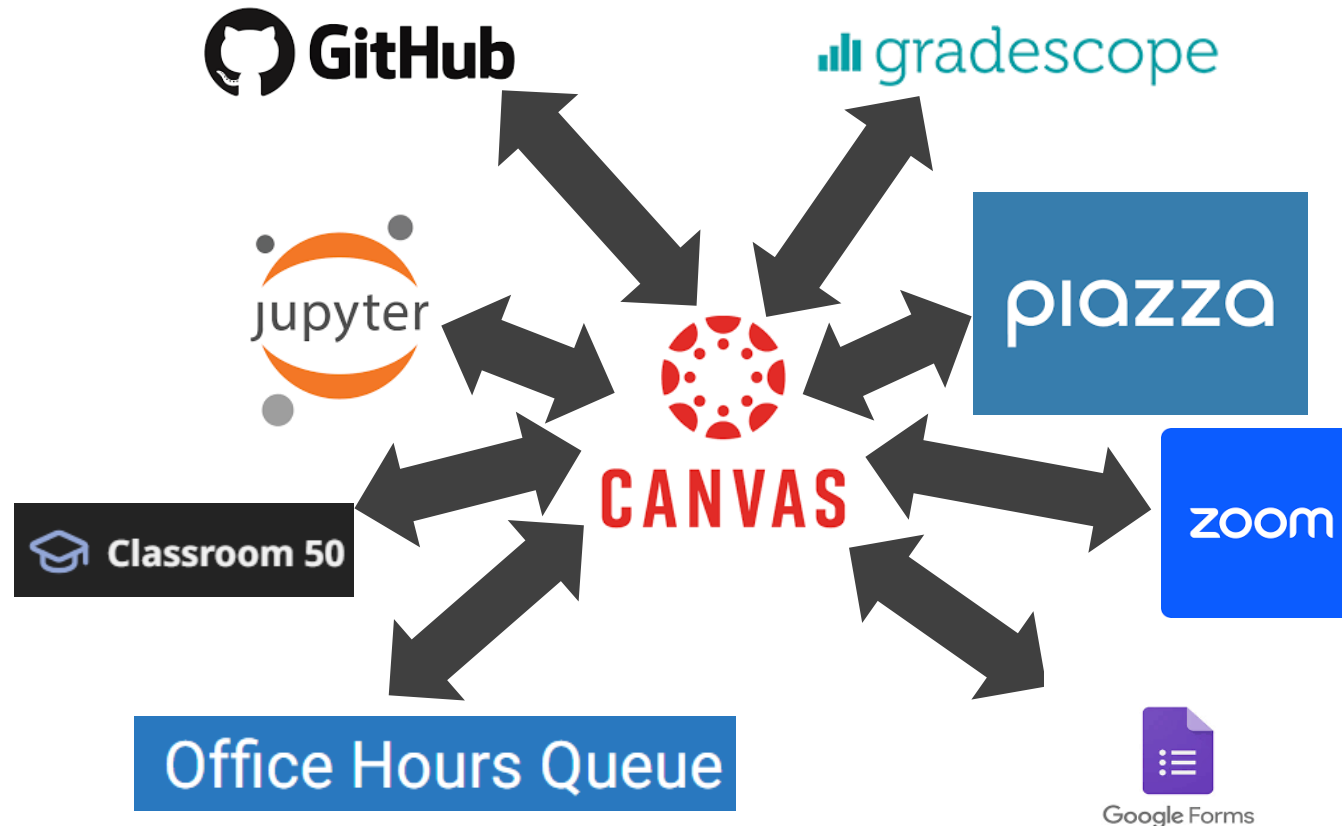
Instructor:

- Mohamed Farag: farag@cmu.edu

TAs:

- Ali Maharramli: amaharra@andrew.cmu.edu
- Yuchen Zhou: yuchenz8@andrew.cmu.edu
- Zihao Du: zihaodu@andrew.cmu.edu

Course Delivery Technology Stack



Course Logistics



- Lectures recordings are made available after the lecture end time.
- You may ask questions interactively or via Google Form during the lecture.
Google Form Link is: <https://forms.gle/P1mQHN9z9JS7fdCJ9> or use the QR code above.
- Use the Student Space Slack Channel to find a teammate for your course project (No instructor or TA help is offered there)
- Students who have approved accommodation shall contact the course instructor to figure out how the instructor can meet their needs.

Course Logistics – Office Hours (Zoom)

OH Schedule (Zoom)							
Days/Timeframes	11am-12pm ET	12-1pm ET	2-3pm ET	4-5pm ET	5-6pm ET	7-8pm ET	9-10pm ET
Monday		Mohamed					Ali
Tuesday	Zihao					Ali	
Wednesday	Yuchen			Zihao			Yuchen
Thursday	Ali			Mohamed	Yuchen		Zihao
Friday			Zihao			Yuchen	
Saturday		Ali					
	Instructor Office Hours - Conducted remotely via Zoom - URL can be found on Canvas						
	TA Office Hours - Conducted remotely via Zoom - URL can be found on Canvas						
	Additional 1-1 TA Office Hours [By appointment only] - Link is shared on Canvas						

- Instructor Office Hours, Recitations and TA Office Hours URLs can be found on Canvas.
- For urgent questions, you can book a 10-minute meeting on Thursday 6-7pm.
 - Spots are limited! Don't overuse this feature.
 - Link to schedule the meeting can be found on Canvas.

Course Logistics – Piazza Hours

Piazza OHs				
	9-10am ET	11am-12pm ET	12-1pm ET	5-6pm ET
Monday	Yuchen			
Tuesday				Zihao
Wednesday			Yuchen	Ali
Thursday	Ali			Zihao
Friday	Yuchen			
Saturday		Zihao		

- Use Course Piazza to ask asynchronous questions that require instructor and/or TA help
- Please note that TAs will respond to inquiries/questions made ***before*** the Piazza OHs start time. Questions and inquiries that are made during the OHs time slot are not guaranteed to be answered during the same time slot.

Office Hours Etiquette Reminder

- Sign-up for a spot in the OH Queue: <https://ohq.eberly.cmu.edu/#/courses> and search for “14-848”
- Office Hours aim to help you find the path to maximize your learning experience.
- Getting the answers from the TA directly won't help you learn so **there won't be direct solutions provided during Office Hours.**
- The goal of the office hours is to **give you some ideas and pointers for you** to debug the issues.
- Please don't plan to spend **more than 15 minutes** in your conversation with the TA.
- Ask **good questions with due diligence**. Please research the issue and put an effort in implementing it before coming to Office hours.
 - **Example of a bad question:** I found this draft code online and I'm citing it but can't get it to work. Can you help?
 - **Example of a good question:** I'm getting a bug in my deployment to the cloud, I researched the issue and found these 3 different references (share the URLs). I implemented the first one and it didn't work. I'm trying the second one now and getting an error that I can't find enough references to it online. What could be the root cause of it?

Course Assessment

Final Exam	Project	Assignments	Quizzes
20%	20%	40%	20%



You can boost your final exam score with up to 4 bonus points



Your lowest quiz score is dropped at the end of the semester
+
2 extra-credit quizzes

Course Assessment – Final Exam on Dec 2 during the lecture

Final Exam	Project	Assignments	Quizzes
20%	20%	40%	20%

Final Exam: is an open-note test.

- Students will have access to all the **PDFs** for lectures, readings and HW solutions. Students can **bring any hard-copied materials with them**.
- Students are required to follow the schedule of their registered section. **On the scheduled final lecture of each section, final exam will be released only to the registered students of the corresponding section.** Each section will have its final exam version(s).
- Exam will be offered via **Lockdown Browser** and **no knowledge exchange is allowed among students during the exam.**
- Students are expected to install and test Lockdown browser on their machines ahead of the exam. If students face an issue with Lockdown browser installation, students must reach out to the instructors **no later than 2 weeks** before the final exam date.
- **Sharing hard-copied notes is prohibited during the exam.**

Course Assessment – Cont'd – Final Exam Booster

Final Exam	Project	Assignments	Quizzes
20%	20%	40%	20%

You can get four bonus points on the final exam if you obtain TWO of the following certifications two weeks before the final exam

- Google Cloud: Associate Cloud Engineer
- AWS Certified Solutions Architect – Associate
- Microsoft Certified: Azure Developer Associate
- GCP Professional Data Engineer
- AWS Certified Data Engineer - Associate
- Microsoft Azure Data Engineer Associate
- Certified Kubernetes Application Developer
- NVIDIA's Fundamentals of Accelerated Data Science

Late submissions are not accepted.

Obtaining only 1 certificate will gain you two bonus points on the final exam score

NVIDIA's workshop is offered in Oct 10.
Interest form is available on Canvas. university

Course Assessment – Cont'd

Final Exam	Project	Assignments	Quizzes
20%	20%	40%	20%

- **Homework Assignments:** there will be 7 homework assignments provided throughout the semester covering the practical aspects of the class and extending the exercises offered during the recitations.
- Students will have 3 days to submit the assignment after the due date with a late penalty. Late penalties are applied based on the timestamp of the last code commit on GitHub and it will follow this equation (no matter whether the delay is in minutes or in hours):
 - Total of 5 points for up to 24 hours delay
 - Total of 15 points for the next 24 hours delay
 - Total of 25 points for the next 24 hours delay
 - 100 points penalty (no grade) after this time.

After homework grades are released, **regrade requests can be made for 24 hours via Gradescope and CANNOT be submitted via email.**

Course Assessment – Cont'd

Final Exam	Project	Assignments	Quizzes
20%	20%	40%	20%

- **Course Project:** Each student will have the option to peer with a team member for the project and you will choose one of three project options to submit. This project leverages most of the topics and practices that are covered throughout the semester. Course project details are released in Week-3.
- **Quizzes:** there will be 1 quiz published on Canvas during the lecture with a specific access code. The access code will be revealed during the lecture to the registered students of the corresponding section.
 - Quizzes will start from the third week of classes.
 - **Quizzes will cover lecture notes, readings, verbal explanations and recitations.**
 - If you encounter extenuating circumstances and must miss a class, please email the course instructor for taking a make-up quiz. Taking the make-up quiz is only available for legitimate extraordinary circumstances (e.g., illness, religious observance, conference attendance, job interviews).



Course Recitations

- Recitations offer important supplementary information, exercises or practices sets that are not covered in class.
- There are 7 recitations held throughout the semester. Refer to the course schedule for the corresponding weeks.
- Typically, recitations are held on Fridays of select weeks. Recitations will use the TA OH URL.
 - TAs may change the recitation date/time with a prior announcement on Piazza.
- Recitations are recorded and will be published on Canvas.
- While live attendance is not required, you may expect quiz questions covering notes presented in the recitations.



Course Grade Scheme

+/- are used to provide granularity in equal intervals in B and C ranges

Grade	Percentage Interval
A/A-	[85-100%], A starts from 93
B	[70-85%)
C	[55-70%)
D	[40-55%)
R (F)	Below 40%

Course Schedule

Preliminary Course Schedule (Subject to Change)

Date	Topic	Notes
Week-1 (Aug 24)	- Introduction & Syllabus - Computing Service Models.	- System Setup homework released
Week-2 (Aug 31)	- Virtualization Basics & Containerization - Lab: Containerization	- System Setup homework deadline. - Docker homework released - Recitation-1: Introduction to Google Cloud
Week-3 (Sep 7)	- Introduction to Agentic Engineering	- Course Project released. - Recitation-2: Docker Scripting and Debugging + Docker Compose Basics.
Week-4 (Sep 14)	- Deployment Orchestration - Pokémon Go Case Study	- Docker homework deadline - Kubernetes homework released. - Recitation-3: YAML Scripting for Kubernetes.
Week-5 (Sep 21)	- Lab: Deployment Orchestration - Kafka	- Recitation-4: More Kafka Examples.
Week-6 (Sep 28)	- Lab: Confluent Kafka - Infrastructure-as-a-Code	- Kubernetes homework deadline - Kafka Homework released - Recitation-5: HCL Basics and Terraform Practice Set.
Week-7 (Oct 5)	- Infrastructure-as-a-Code (Cont'd) Terraform - Lab: Terraform	- Course Project Checkpoint

Course Schedule – Cont'd

Week-8 (Oct 19)	- Cloud Infrastructure Concepts	- Kafka Homework deadline - Terraform Homework released
Week-9 (Oct 26)	- Introduction to Hadoop	- Terraform Homework deadline
Week-10 (Nov 2)	- Hadoop HDFS - Hadoop MapReduce	- Hadoop MapReduce homework released
Week-11 (Nov 9)	- Big Data Algorithms - Introduction to Spark	- Hadoop MapReduce homework deadline - Recitation-6: Hadoop MapReduce Practice Set.
Week-12 (Nov 16)	- Spark (Cont'd) - Lab: Spark Programming	- Course project submission deadline - Apache Spark homework released - Recitation-7: Spark Practice Set.
Week-13 (Nov 23)	- Cloud Security & DevSecOps	- Apache Spark homework deadline
Week-14 (Nov 30)	- Final Exam Review - Final Exam	



Expectations down the Road!

- For the final exam, we will use the Lockdown browser, and you will have access to the lecture PDFs via the browser. You will receive few training attempts on the final exam environment before the exam date.
- You are welcome to use other cloud platforms like Azure and AWS. That said, we don't have any credits/support that we can offer for these systems. You may need to use a Free Trial version.

HW Submission Guidelines

- HW-1 will focus on Environment Setup and essential GitHub Skills.

- HW-1 submission:



- HW-2 and later assignment submissions:





Academic Integrity Violations (AIVs)

- AIVs are serious and can have direct impact on your course grade, your scholarship -if any-, your graduation timeline, and/or your continuation in your degree program.
- You may use all the materials and readings that we explicitly provide you with in the assignment document. In this case, attribution/citation is not required.
- You may use other material from the course, including lectures, Piazza posts by the instructors. For any such use, you must provide a proper attribution.
- You may not obtain code or other solution information from an unauthorized external source, including web pages, code repositories, blog posts, etc.
- You may not look at someone else's code (or other documents.) This includes one person looking at code and describing it to another. There is no notion of looking “too much,” since no looking is allowed at all.



Academic Integrity Violations (AIVs) – Cont'd

- For each assignment, the instructor may randomly invite a group of students to conduct oral discussions to learn how the students implemented the assignment. Students may face academic integrity violations if they fail to explain their own work. However, students can't use the direct homework question in their prompt to the AI tool.
- Students cannot publish their assignment solutions or project work publicly without an explicit approval from the instructor.
- Students are not allowed to share Quiz Access Code with their peers who didn't attend the class in person without pre-approval from the instructor. Sharing the quiz access code, without instructor's approval, is considered an academic integrity violation.



Our View on AI Tools

- AI can be a productive tool to your workflow. However, it can also be a diminisher to your critical skill abilities.
- In this course, our approach is **UBAS**.
- UBAS stands for: Understand the required concept(s), Build the necessary critical thinking abilities, Apply those abilities in a practical setting, Sstreamline your workflow, when appropriate, using AI tools.
- Students will have pre-identified tasks where they can use GenAI tools (including coding agents and chatbots). Generally, the use of GenAI tools is prohibited unless explicitly stated in the homework document. If explicitly stated, students can use GenAI tools and attribute the work generated by AI properly.

Why Not Overuse AI?

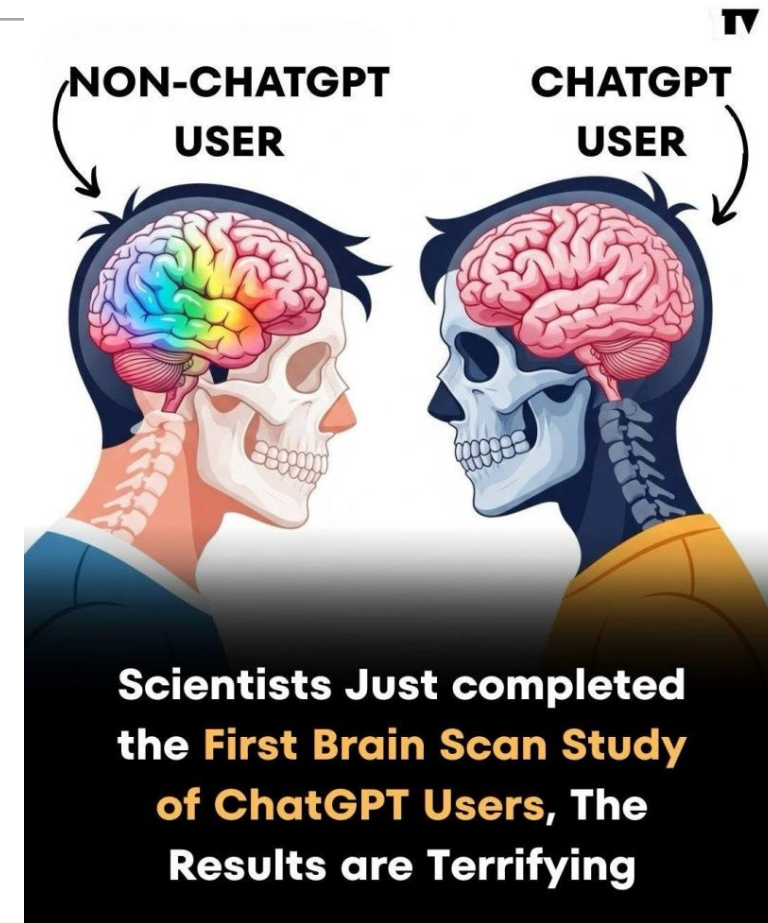
1. Concerns on the development of your basic cognitive skills.
2. Concerns on your creativity abilities.

AI in Writing Essays

From “Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task”:

<https://www.media.mit.edu/publications/your-brain-on-chatgpt/>

- Study compared three groups writing essays: LLM-assisted, Search Engine assisted, and Brain-only (no tools), over multiple sessions.
- Used EEG to measure brain connectivity; human and AI judges also scored essays.



Carnegie Mellon University

AI in Writing Essays – Findings

From “Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task”:

<https://www.media.mit.edu/publications/your-brain-on-chatgpt/>

- LLM assistance group suffered reduced neural connectivity, particularly weaker distributed networks, lower engagement relative to other groups.
- Over time, LLM users showed lower memory recall and more under-engagement; in swapped-to-brain-only condition, some reduced connectivity persisted.
- LLM group reported lowest sense of essay ownership; they had more difficulty accurately quoting their own writing.
- **Overusing LLM tools may bring cognitive costs, reduced mental effort, and engagement.**

AI in Ideation

From “The Ideation-Execution Gap: Execution Outcomes of LLM-Generated versus Human Research Ideas”: <https://arxiv.org/pdf/2506.20803>

- Generative AI generate research ideas that rate, initially, higher than human-expert ideas at ideation stage (novelty, excitement, expected effectiveness, overall quality).
- When these ideas are executed by expert implementers, the LLM ideas' scores drop much more than human ideas on all key metrics (novelty, excitement, effectiveness, overall quality).
- This gap (“ideation-execution gap”) is statistically significant: human ideas stay more stable; LLM ideas lose more of their initial promise.
- **Idea execution reveals factors often missed at ideations by LLMs such as empirical performance, rigorous experiment design, baseline comparisons, feasibility and resource constraints.**



Teaching Expectations

- Not all material is captured on the slides; students are expected to take notes during lectures.
- Students are expected to attend recitations (or watch the recordings) prior to the following lecture.
- Students are expected to complete all assigned readings and in-class exercises on-time.
- Instruction is delivered in a passionate teaching style. Don't be scared! 😊

Other Syllabus Information

- **If you run out of Google Cloud credits, plan for 24-48-hour delay to get a new coupon.**
- Syllabus contains important information about student wellness, student academic success center, and food insecurity.
- The Syllabus can be found on Canvas under the Modules section

Waitlisted?

For enrollment questions and inquiries, please
email the INI Academics at

ini-academic@andrew.cmu.edu



Remember!

- You are a graduate student at CMU, and we expect you to **pay close attention to the details** mentioned during lectures and in your homework assignments.
- Ask questions and **avoid making unreasonable assumptions**.
- Use your **intellectual abilities and problem-solving skills** to fill in any gaps beyond what the teaching team can share with you.
- We are **committed to your success** both inside and outside the classroom.
- Feel free to **reach out with any questions** related to research, internships, career advice, or related topics.

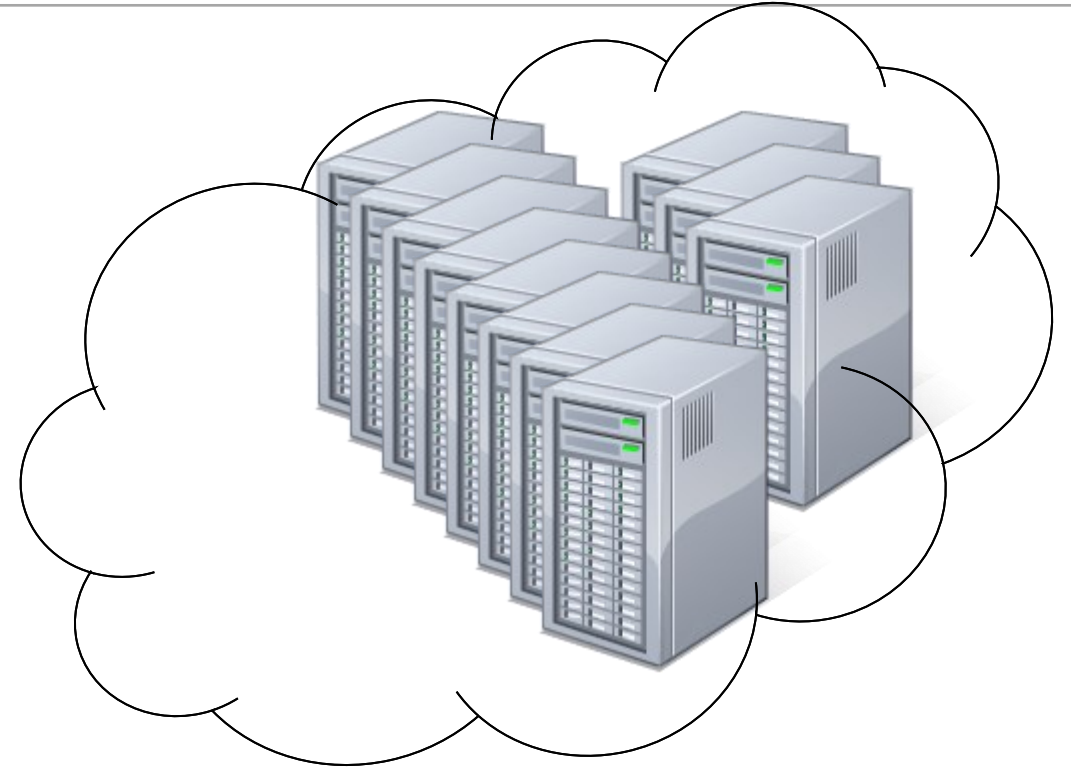


Now, let's start!
What is the Cloud?

What is the Cloud?



- Is it a cluster?
- Is it a supercomputer?
- Is it a datastore?



- In simple words:
 - Cloud = lots of **S**Storage + **C**Compute cycles nearby + **N**Network bandwidth

Scientifically, what is the Cloud?

Cloud is a model that offers the following characteristics:



On-demand
self-service

No human
intervention
needed to get
resources



Broad network
access

Access
from
anywhere



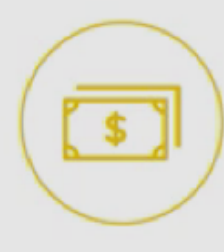
Resource
pooling

Provider
shares
resources
to
customers



Rapid
elasticity

Get more
resources
quickly as
needed



Measured
service

Pay only
for what
you
consume

Neocloud: AI-first Cloud

- Neocloud is a specialized cloud built for AI and high-performance computing.
- For example, GCP is designed so you can host a website, run PostgreSQL, store files in buckets, run Kubernetes, and rent GPUs.
 - A neocloud is saying: “We specialize in giving you hundreds or thousands of GPUs connected together efficiently so you can train or run AI models.”

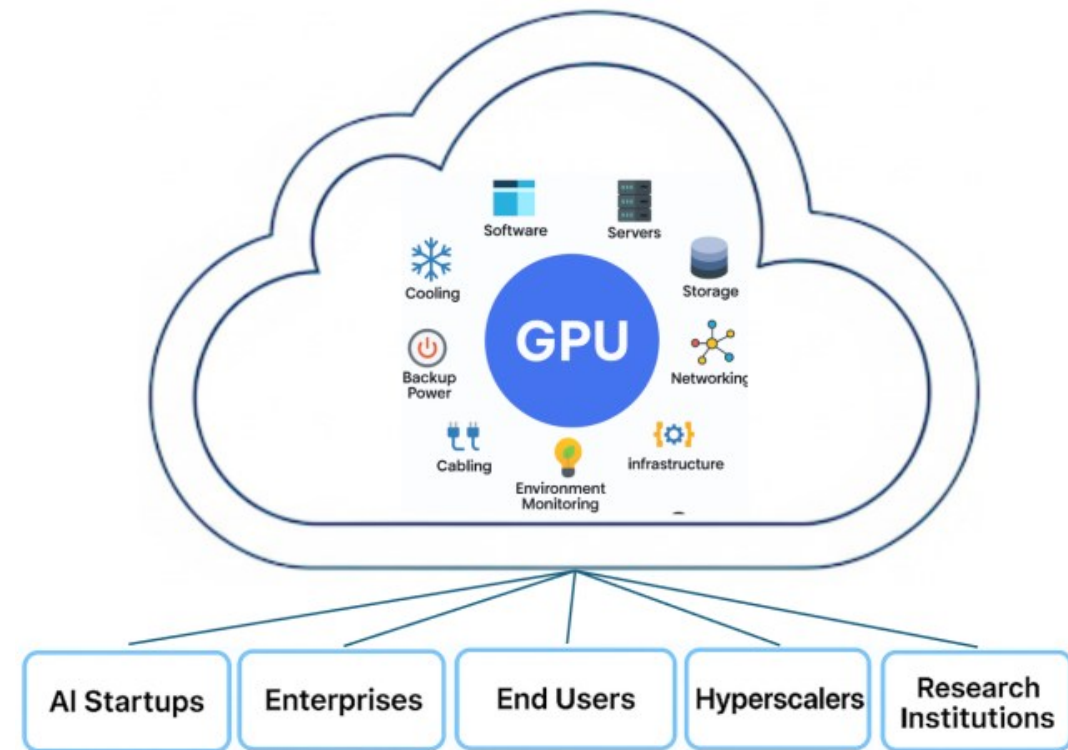


Image from DataRaAI

Why Do We Care about Cloud Infrastructure?



Accessibility



Big Data



Cost

Examples of Cloud Services

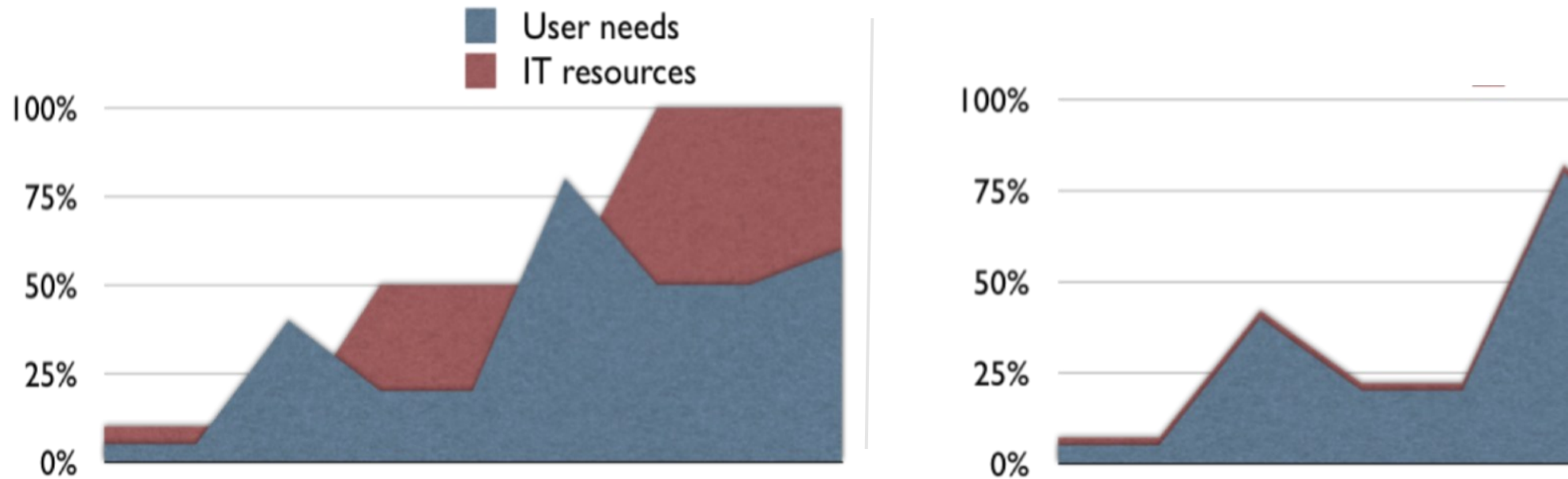
- Dropbox
- Google Drive
- Microsoft OneDrive
- Apple iCloud



- Netflix - hosted on AWS
- Google search – Google Cloud
- Google Docs, Sheets, and Slides



Company Infrastructure and User Needs With vs. Without Cloud



**Which diagram reflects the IT resources and User needs for Infrastructure using the Cloud?
And Why?**



Conventional Computing Infrastructure vs Cloud Computing Infrastructure

Conventional

Manually Provisioned
Dedicated Hardware
Fixed Capacity
Pay for Capacity
Capital & Operational Expenses
Managed via System administrators

Cloud

Self-provisioned
Shared Hardware
Elastic Capacity
Pay per Use
Operational Expenses
Managed via APIs

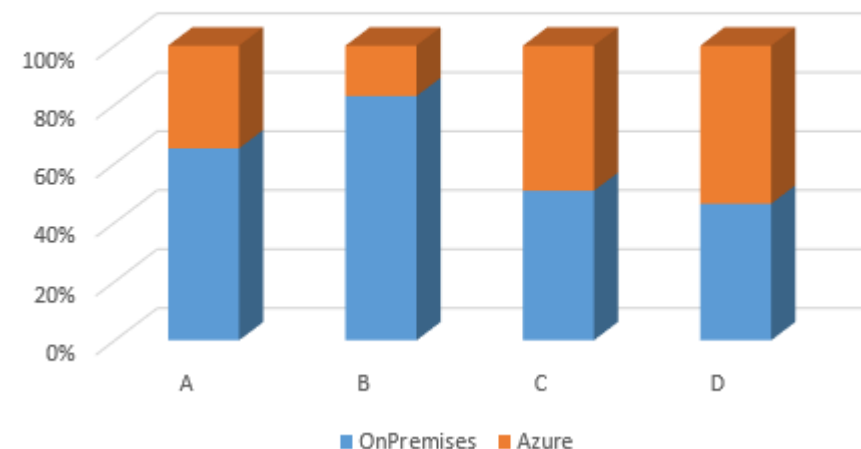
Cloud Cost Savings by Deployment Size

Consider the following scenarios for resource deployment:

- A. Modest Deployment
- B. Tiny Deployment
- C. Enterprise
- D. Very Large Enterprise

Scenario	A	B	C	D
Small VM	20	4	200	500
Medium VM	40	10	750	1750
Large VM	10	2	150	750
Storage (TB)	10.9	2.5	171.9	468.8

Costs Compared





Why the cloud is more economical option for tiny and modest deployments?

Aspect	Traditional IT Costs	Cloud-Based Server Costs
Initial Setup Costs	High, includes hardware, software, and infrastructure setup	Low to moderate, primarily subscription-based or pay-as-you-go
Maintenance and upgrades	High, ongoing costs for hardware maintenance and software upgrades	Low, handled by cloud service providers
Scalability	Limited, requires significant investment for expansion	High, easily scalable based on demand
Operational flexibility	Low, fixed resources and limited flexibility	High resources can be adjusted as needed
Disaster recovery	High, requires investment in backup infrastructure	Low to moderate, often included in cloud services
Energy and cooling costs	High, significant energy consumption and cooling requirements	Low, managed by cloud providers
Staffing requirements	High need for IT staff to manage and maintain systems	Low, reduced need for in-house IT staff
Downtime costs	Potentially high, dependent on in-house capabilities	Low, high availability and reliability offered by providers
Security costs	High, requires investment in security measures	Variable, often included in cloud service plans
Time to deployment	Long, extensive setup and configuration time	Short, rapid deployment and provisioning of resources



Cloud Infrastructure Enablers

Data Center + Virtualization
(Hardware) (Software)



Next Steps

- Complete Course Entry Survey
 - Link can be found on Canvas under “Modules” section.
- Join the Course Piazza.
- Join the student Slack workspace.
- Read the Course Syllabus.

Waitlisted Students

- All materials for first two weeks will be uploaded here

